Electron-Proton Path Integral Monte Carlo with the Pair-Product Action

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We study the electron-proton binary mixture with the path integral Monte Carlo method. In order to lower the discretization error in the approach to the continuum limit we use the (end-point approximation to the) pair-product Coulomb inter-action instead of the primitive approximation and a mix of the two keeping the former for the electron-proton component only and the latter for the like components. We determine some thermodynamic properties like the internal energy and the superfluid fractions and structural properties like the partial and total radial distribution functions. We compare the realistic mixture where the like particles obey to the Fermi-Dirac statistics with the artificial one where the particles obey to the Bose-Einstein statistics. While for bosons the method is exact for fermions we employ the approximate restricted path integral method with a free particle restriction. We study the mixture in its hot dense metallic atomic Hydrogen phase. We discuss the importance of approaching the continuum limit.

Keywords: Path Integral; Monte Carlo; Quantum Mixture; Bosons; Fermions; Two Component Plasma; Thermodynamics; Structure; Superfluidity; Sign Problem; Metallic Hydrogen; Molecular Hydrogen; Hydrogen Atom; Hydrogen Molecule; Pair Coulomb Density Matrix; Continuum Limit

CONTENTS

I. INTRODUCTION

Hydrogen is the simplest element in the Universe and the most abundant in nature. On general grounds we expect a plasma of electrons and protons (*ep* plasma) to form clusters of one electron bound to one proton, i.e. Hydrogen atoms. Therefore we expect that a computer experimenter which simulates the *ep* plasma in thermal equilibrium will observe the H atom formation. That this is possible using just the laws of quantum statistical physics and the law of Coulomb has been the subject of many studies starting from the pioneer work of Edwards and Lenard [1, 2]. In this work we worry about the particular many body Monte Carlo [3] simulations. There is an extensive literature on first principle computational studies of the *ep* plasma as a fluid binary mixture [4]. In this work we use the particular path integral Monte Carlo (PIMC) method [5].

If we were to study the non-quantum *ep* binary mixture (also called a two component plasma [6]) we would need to introduce some hard core for the opposite charge particles not to stick one on the other due to the Coulomb attraction. This is not necessary if we start from a quantum description since a pair of charges in the plasma must satisfy the cusp condition [7, 8] which prevents the divergence of the pair interaction at contact. From a path integral numerical computation point of view, taking properly into account of the cusp condition allows to calculate the path

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integral, discretized in the imaginary time, with much less effort than using the bare Coulomb interaction, since it makes possible to choose a much less dense time discretization when approximating the continuum limit. We will compare two scenarios: i. using the end-point approximation in the pair-product Coulomb inter-action [5] both for the like and unlike components; ii. using the end-point approximation in the pair-product Coulomb inter-action [5] only for unlike component, which is a necessary condition in order to work without hard cores.

Both the electron and the proton are spin 1/2 particles. Moreover we will only consider the case of polarized electrons and protons, so that both species of the *ep* mixture, the identical many electrons and the identical many protons, must obey to Fermi-Dirac statistics. This is responsible for the so called 'exchange hole' in the close to contact structure of each species of the mixture through the Pauli exclusion principle. But like particles also repel each other due to the Coulomb repulsion. We will refer to this effect as the 'correlation hole' in the close to contact structure of each species so that we can say that the close to contact structure of each species is affected by the 'exchange-correlation hole'.

In this work we will always compare the realistic case that treats separately the two species as fermions with the artificial case of treating them as bosons. We do this because while the simulations for bosons are exact, the simulations for fermions are affected by the so called 'fermions sign problem' [9]. So the simulations of the realistic case can only be treated with approximate methods. We will here use the restricted path integral Monte Carlo (RPIMC) method with a free particles restriction [10]. The RPIMC method is much more demanding computationally since one has to calculate determinant of order of the number of the particles of each species, $N/2$, for each timeslice of the time discretization, after each Metropolis [11] move.

We expect a phase transition from a classical TCP at high temperature, $T > 10^4$K, to metallic Hydrogen, $10^3\text{K} < T < 10^4$K, to molecular Hydrogen, $10\text{K} < T < 10^3$K, to solid Hydrogen at lower temperatures (see figure 4 of Ref. [4] or figure 2 of Ref. [12]). A liquid-liquid phase transition at approximately 500GPa separates the molecular Hydrogen at lower pressures to the atomic Hydrogen at higher pressures. In this work we will use the canonical N, n, T ensemble where n is the number density and will simulate just the metallic atomic phase and the incipient molecular phase. Since we used polarized protons, the molecular phase expected at lower temperatures and/or lower densities would be that of orthohydrogen where the nuclei of the single H_2 molecules are in a triplet state.

We will calculate numerically thermodynamic quantities like the total kinetic and potential energy and the superfluid fractions for each species. And structural quantities like the partial and total radial distribution function. Our computer experiments will simulate either a planet interior like the one for Jupiter [13] or a earthly laboratory conditions like the one of high pressures under diamond anvil cells [14].

II. THE COMPUTER EXPERIMENT

In order to study the quantum statistical properties of the *ep* fluid binary mixture (H) through a computer experiment it is necessary to add to our simulation box of volume Ω an equal number $N/2$ of electrons (e) of mass m_e, the species 1, and of protons (p) of mass m_p, the species 2, and let them interact with the usual Coulomb pair potential $\pm 1/r$. We thus obtain a fluid of number density $n = N/\Omega$. We can then study the properties of H in thermal equilibrium at an inverse temperature $\beta = 1/k_BT$ with k_B Boltzmann constant and T the absolute temperature. If the laws of quantum statistical physics and of Coulomb are meaningful we should be able to reproduce the thermodynamic and structural properties of H that we observe in nature.

In our reduced units we will measure lengths in units of a Bohr radius $a_B = \hbar^2/m_e e^2 \approx 0.529$Å and energies in units of two Rydbergs $e^2/a_B \approx 27.2$eV, i.e. a Hartree. The Hamiltonian operator and the density matrix in reduced units will then read

$$H = K + V = -\sum_{\alpha=1}^{N} \lambda_{i_\alpha} \nabla_\alpha^2 + \sum_{\alpha<\beta=1}^{N} \phi_{i_\alpha i_\beta}(r_{\alpha\beta}), \tag{2.1}$$

$$\rho = e^{-H/T}. \tag{2.2}$$

where we indicate with a Greek index the particle label and with a Roman index the species label so that for example i_α is the species label of particle α, either 1 for the electrons or 2 for the protons, $\boldsymbol{r}_\alpha$ is the position vector of particle α, and $r_{\alpha\beta} = |\boldsymbol{r}_{\alpha\beta}| = |\boldsymbol{r}_\alpha - \boldsymbol{r}_\beta|$ is the distance between particles α and β. The partial pair potential is the usual Coulomb potential $\phi_{ij}(r) = \epsilon_{ij}/r$ with $\epsilon_{ii} = +1$ for all i and $\epsilon_{ij} = -1$ for $i \neq j$. $\lambda_i = \hbar^2/2\mu_i$ where the reduced masses are $\mu_1 = 1$ and $\mu_2 = m_p/m_e \approx 1836.15$. Temperatures are in units of $\mathscr{T} \approx 315775$K.

The coordinate representation of the high temperature density matrix for distinguishable particles is [5]

$$\rho(R, R'; \tau) = \rho_0(R, R'; \tau) e^{-U(R,R';\tau)}, \tag{2.3}$$

$$\rho_0(R, R'; \tau) = \prod_{\alpha=1}^{N} (4\pi\lambda_{i_\alpha}\tau)^{-3/2} e^{-(\boldsymbol{r}_\alpha - \boldsymbol{r}'_\alpha)^2/4\lambda_{i_\alpha}\tau}, \tag{2.4}$$

where ρ_0 is the high temperature free particles density matrix with $\tau = \beta/M$ for large M and U is the *inter-action*. So that the coordinate representation of the thermal density matrix for identical particles can be obtained from the following discretized path integral [10]

$$\rho(R, R'; \beta) = \sum_{\mathscr{P}} \text{sgn}(\mathscr{P}) \int \rho(\mathscr{P}R, R_1; \tau)\rho(R_1, R_2; \tau) \cdots \rho(R_{M-1}, R'; \tau) \prod_{k=1}^{M-1} dR_k, \tag{2.5}$$

where $R_k = (\boldsymbol{r}_{1,k}, \boldsymbol{r}_{2,k}, \ldots, \boldsymbol{r}_{N,k})$ is the kth timeslice. Here $\boldsymbol{r}_{\alpha,k}$ is the position vector of particle α of species i_α at timeslice k in the path integral. Each high temperature density matrix gives a *link* between two successive *timeslices*. The sum is over the permutations $\mathscr{P}$ of identical particles (of the same species) and sgn($\mathscr{P}$) is the sign of the given permutation

$$\text{sgn}(\mathscr{P}) = \begin{cases} +1 & \text{Bose-Einstein statistics} \\ \begin{cases} +1 & \mathscr{P} \text{ an even permutation} \\ -1 & \mathscr{P} \text{ an odd permutation} \end{cases} & \text{Fermi-Dirac statistics} \end{cases} \tag{2.6}$$

While a Monte Carlo integration [3] of the path integral (2.5) (PIMC) can be carried out exactly for bosons, for fermions the alternating signs in the permutation sum becomes less and less efficient as N and/or M increase [9]. This is known as the 'fermions sign problem'. An approximate way to circumvent this problem is the Restricted Path Integral Monte Carlo (RPIMC) method of Ceperley [9, 10]. In our simulations we will always compare the PIMC for bosons and the RPIMC for fermions with a free fermions restriction. Of course the Bose-Einstein case is not realistic but we offer it just as a theoretical and exact calculation.

The *primitive approximation* to the inter-action amounts to the simple choice $U(R, R'; \tau) = -\tau[V(R) + V(R')]/2$ but is subject to a large discretization error [5]. In order to reduce the discretization error we will here adopt the two following choices for the inter-action U:

i. the end-point approximation for the pair-product action (see section IV.F of Ref. [5]) on all three components ee, pp, and ep,

$$U_{\mathscr{A}}(R, R'; \tau) \approx \sum_{\alpha<\beta} P_{i_\alpha i_\beta}(\boldsymbol{r}_{\alpha\beta}, \boldsymbol{r}_{\alpha\beta}; \tau), \tag{2.7}$$

where for P_{ij} we choose the pair Coulomb density matrix of Pollock [8]. Namely

$$P_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau) = -\ln\left[\frac{\rho_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau)}{\rho^0_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau)}\right], \tag{2.8}$$

where

$$\frac{\partial \rho_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau)}{\partial \tau} = \left[\lambda_{ij}\boldsymbol{\nabla} - \frac{\epsilon_{ij}}{r}\right] \rho_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau). \tag{2.9}$$

Here $\lambda_{ij} = \lambda_i + \lambda_j$ and $\rho^0_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau) = (4\pi\lambda_{ij}\tau)^{-3/2} \exp[-(\boldsymbol{r} - \boldsymbol{r}')^2/4\lambda_{ij}\tau]$ satisfies the same Eq. (2.9) but for $\epsilon_{ij} = 0$ for all i, j;

ii. the end-point approximation for the pair-product action only on the unlike ep component and the bare Coulomb pair-potential on the others two like components,

$$U_{\mathscr{B}}(R, R'; \tau) \approx \sum_{\alpha<\beta, i_\alpha \neq i_\beta} P_{i_\alpha i_\beta}(\boldsymbol{r}_{\alpha\beta}, \boldsymbol{r}_{\alpha\beta}; \tau) - \tau \sum_{\alpha<\beta, i_\alpha = i_\beta} \frac{\epsilon_{i_\alpha i_\beta}}{r_{\alpha\beta}}. \tag{2.10}$$

In a PIMC computer experiment one needs to worry about two limiting procedures: i. the thermodynamic $N \to \infty$ limit and ii. the continuum $M \to \infty$ limit. The simulation box of volume $\Omega = L^3$ can be made periodic so that it permeates the whole infinite three dimensional space in order to mimic the thermodynamic limit. This will produce finite size errors that can be reduced by increasing the box size L. Of course in order to study a given thermodynamic state of the system this requires increasing L at constant density n which means increasing N. This will make the computer experiment more demanding from the numerical calculation cost point of view. Moreover, in periodic boundary conditions the potential energy V for the particles within the central box should include also the pair interactions with all their images in the periodic boxes. Since the Coulomb potential $1/r$ is long range these contributions cannot generally be neglected and specific techniques should be used [15] to treat them, the most common of which is the Ewald summation. In this work we will neglect interactions with periodic images and we use the end-point approximation for the pair-product action. The continuum limit requires an M sufficiently large for the characteristic size of a free link of each species be much smaller than the box side, i.e. $\sigma_i = \sqrt{2\lambda_i\tau} \ll L$ for all i, so that we can neglect the periodic copies of the high temperature free density matrix.

It is often useful to introduce the Wigner-Seitz radius r_s for the electron component, i.e. the reduced mean interelectron distance $r_s = (4\pi n_1/3)^{-1/3}$ with $n_1 = n/2$. If we rescale each reduced coordinate as $\tilde{\boldsymbol{r}}_\alpha = \boldsymbol{r}_\alpha/r_s$. Then the rescaled kinetic energy, $\tilde{K} = r_s^2 K$ and the rescaled potential energy, $\tilde{V} = r_s V$. We then see that as n increases, r_s decreases, V is penalized respect to K, and the mixture tends to behave like a non interacting one, with less correlations. This is confirmed by our numerical experiments.

We performed some computer experiments as described in Table I. During the simulations we measured some thermodynamic quantities as the total kinetic and potential energy of the ep plasma and the superfluid fractions for each species. These were measured using the thermodynamic estimators described in Ref. [5]. And we also calculated the partial, $g_{ij}(r)$, and total, $g_{\rm tot}(r) = [g_{11}(r) + 2g_{12}(r) + g_{22}(r)]/4$, Radial Distribution Functions (RDF) [16]. Given an observable $\mathscr{O}$ we can measure it during the simulation through the following PIMC operation [5]

$$\langle \mathscr{O} \rangle = \mathrm{tr}(\rho\mathscr{O})/Z, \tag{2.11}$$

$$Z = \mathrm{tr}(\rho), \tag{2.12}$$

where ρ is the density matrix of Eq. (2.5), $\mathrm{tr}(\ldots)$ is the trace operation, and Z is the canonical partition function. Therefore we impose both spatial and imaginary time periodic boundary conditions for the many body path of both species $R(t) = (\boldsymbol{r}_1(t), \boldsymbol{r}_2(t), \ldots, \boldsymbol{r}_N(t))$ where $t \in [0, \beta[$ is the imaginary time. So that for any function f we require $f(R + L\boldsymbol{u}) = f(R)$ with $\boldsymbol{u}$ a unit vector along any of the $3N$ dimensions and $R(t+\beta) = R(t)$. Our algorithm is discussed in the Appendix A.

For each thermodynamic state considered we compared the two cases where we treat the like particles first as bosons and then as fermions. Comparing the reduced de Broglie thermal wavelength of each species, $\Lambda_i = \sqrt{1/T\mu_i}$, with the interparticle spacing, $(n/2)^{-1/3}$, we may define a reduced degeneracy temperature, $T_i^D = (n/2)^{2/3}/\mu_i$, for each species. For temperatures higher than the degeneracy temperature, quantum statistics, either bosonic or fermionic, are not very important. Note that the difference between the two statistics for the heavy and slow protons component is driven by the light and fast electrons component which 'dress' the protons.

In Fig. 1 we compare the end-point approximation for the pair-product action [5] constructed with the pair Coulomb density matrix of Pollock [8] with the bare Coulomb potential that enters the primitive approximation to the inter-action. The figure shows how $P_{ij}(\boldsymbol{r}, \boldsymbol{r}; \tau)$ from the pair Coulomb density matrix (2.8) does not diverge to $\pm\infty$ for $r \to 0$ but satisfies a cusp condition instead. The cusp condition can be easily determined studying the asymptotic behavior at small r of a vanishing residual energy for the pair-product approximation to the inter-action [5]. Clearly this is a purely quantum effect and it allows to avoid the use of an artificial hard core for the proton species.

In Figs. 2,3 and 4,5 we show the RDF's for the high temperature case studies of Table I. From the like RDF we see how the ee component has the expected exchange hole near contact, $r \to 0$, due to Pauli exclusion principle affecting the polarized electrons. An important difference between the $\mathscr{A}$ and the $\mathscr{B}$ cases is that the contact value for $g_{ee}(0)$ is zero only in the $\mathscr{B}$ case. In the $\mathscr{A}$ case the approximate free particle restriction of the RPIMC (see Appendix A) is not enough to win over the tendency of the electrons to pile up close to the protons. The ep component has a high peak at contact which signals the H atom formation. The pp component has a first peak at $r \approx 1.5$ in the bosons case which is washed out in the fermions case. As the density is lowered at a constant low temperature, the peaks increase in magnitude and the difference between the Bose and Fermi cases increases, as expected. In these cases the real mixture is still in a metallic Hydrogen phase and our results for the $\mathscr{B}$ case compare well with the study of Ref. [17].

In order to simulate the interior of planets like Jupiter or Saturn we should be able to treat reduced temperature as low as $T \approx 10^4/\mathscr{T} \approx 0.03$ or less. To reach these temperatures keeping $\tau = 0.3$ constant requires $M \gtrsim 100$ (lowering M will bring us farther away from the continuum limit and the potential energy of the mixture will become more negative). We then considered cases E and F of Table I. The RDF's are shown in Figs. 6 for case $\mathscr{A}$ and in 7 for case $\mathscr{B}$. We see that the behavior of the fermionic real mixture is profoundly different from that of the bosonic mixture.

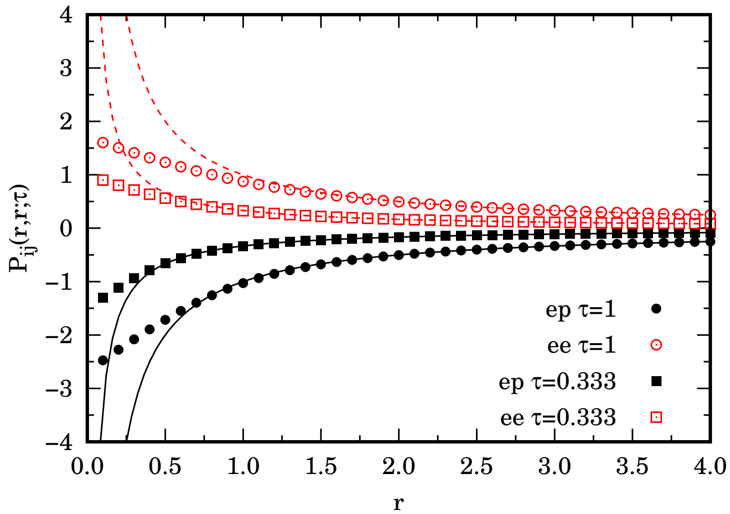

FIG. 1. We show the ee and ep components of $P_{ij}(\boldsymbol{r}, \boldsymbol{r}; \tau)$ of Eq. (2.8) for two values of τ (dots). We compare it with the bare Coulomb potential $\tau\epsilon_{ij}/r$ (lines). From the figure we can see that the cusp condition [8] $\lim_{r\to 0} dP_{ij}(r, r; \tau)/dr \sim -\epsilon_{ij}/\lambda_{ij}$ is satisfied at small r. The distance r is in units of a Bohr radius.

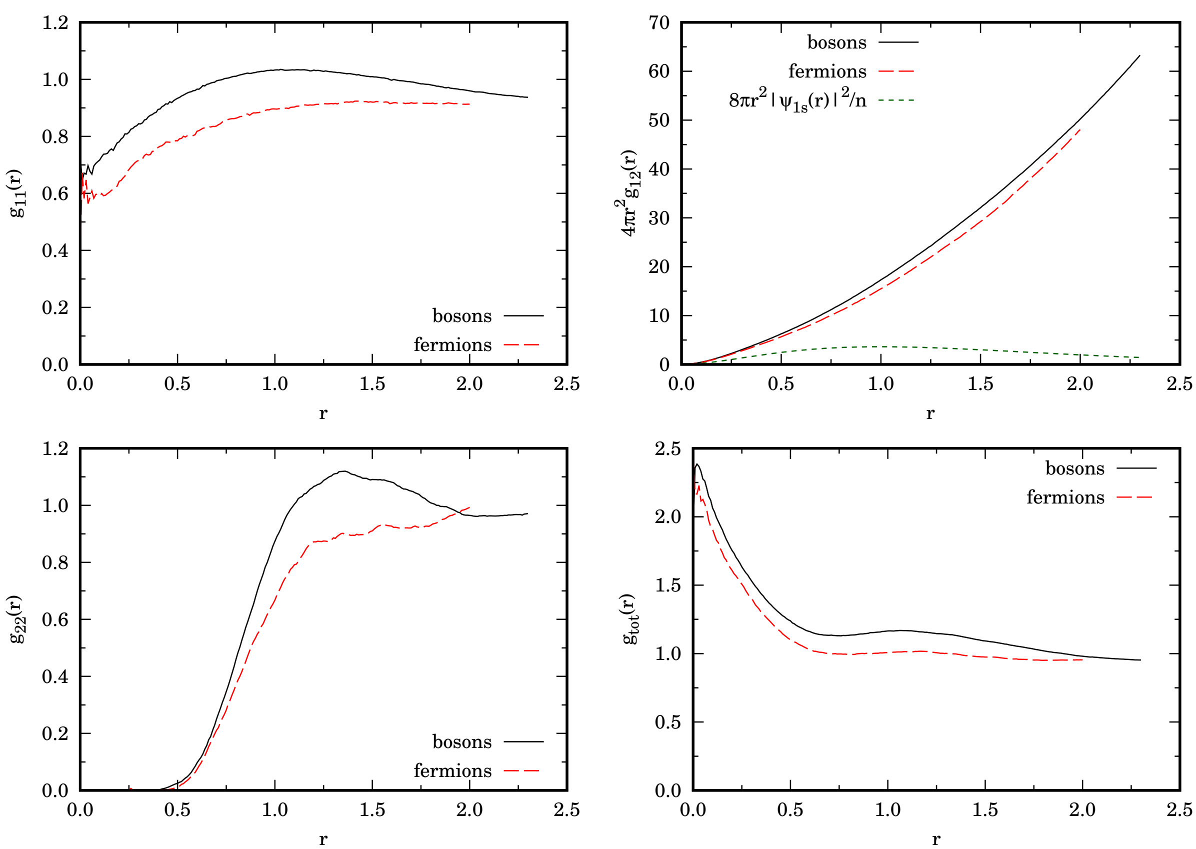

FIG. 2. We show the partial and total RDF of the H fluid for cases A$\mathscr{A}$ and B$\mathscr{A}$ in Table I where $\tau = 0.333$. In the panel for the unlike radial distribution function we also show the ground state probability distribution of the H atom $|\psi_{1s}|^2$ for comparison. The distance r is in units of a Bohr radius.

TABLE I. Cases treated in our simulations for the ep fluid mixture. The calligraphic upper case letter indicates the choice of the inter-action: $\mathscr{A}$ for (2.7) and $\mathscr{B}$ for (2.10). We show the total number of particles N, the reduced number density $n = N/\Omega$, the reduced temperature T, the total kinetic energy per particle $e_K = \langle K\rangle/N$, the total potential energy per particle $e_V = \langle V\rangle/N$, the superfluid fraction for species i, $f_s(i)$. In all cases $\tau = 1/TM \approx 0.3$. In the statistics column "b" stands for bosons, treated with the exact PIMC, "f" for fermions, treated with the approximate RPIMC. The simulations lasted no less than 10^8 MC steps. The simulation of case F lasted for about two months of computer time.

inter-action	case	statistics	T	M	n	r_s	N	Ne_K	$-Ne_V$	$f_s(1)$	$f_s(2)$
	A$\mathscr{A}$	b	0.15	20	0.3	1.17	30	10.44(1)	20.88(1)	0.596(1)	0.04774(7)
	B$\mathscr{A}$	f	0.15	20	0.3	1.17	20	6.2(2)	13.83(5)	0.72(6)	0.0475(3)
$\mathscr{A}$	C$\mathscr{A}$	b	0.15	20	0.16	1.44	30	9.83(1)	19.55(2)	0.486(2)	0.04783(7)
	D$\mathscr{A}$	f	0.15	20	0.16	1.44	20	5.4(1)	12.06(5)	0.64(5)	0.0477(3)
	E$\mathscr{A}$	b	0.03	111	0.16	1.44	30	17.58(2)	37.90(3)	0.0164(2)	0.00898(3)
	F$\mathscr{A}$	f	0.03	111	0.16	1.44	20	3.27(5)	11.55(1)	0.74(4)	0.00904(5)
	A$\mathscr{B}$	b	0.15	20	0.3	1.17	30	10.19(1)	20.072(9)	0.618(1)	0.04745(6)
	B$\mathscr{B}$	f	0.15	20	0.3	1.17	20	6.90(5)	13.25(2)	0.68(2)	0.0478(6)
$\mathscr{B}$	C$\mathscr{B}$	b	0.15	20	0.16	1.44	30	9.04(1)	17.82(1)	0.581(2)	0.04752(7)
	D$\mathscr{B}$	f	0.15	20	0.16	1.44	20	5.65(5)	11.62(2)	0.66(2)	0.048(1)
	E$\mathscr{B}$	b	0.03	111	0.16	1.44	30	9.24(1)	24.19(2)	0.185(1)	0.00895(3)
	F$\mathscr{B}$	f	0.03	111	0.16	1.44	20	3.66(4)	11.62(1)	0.63(3)	0.00895(5)

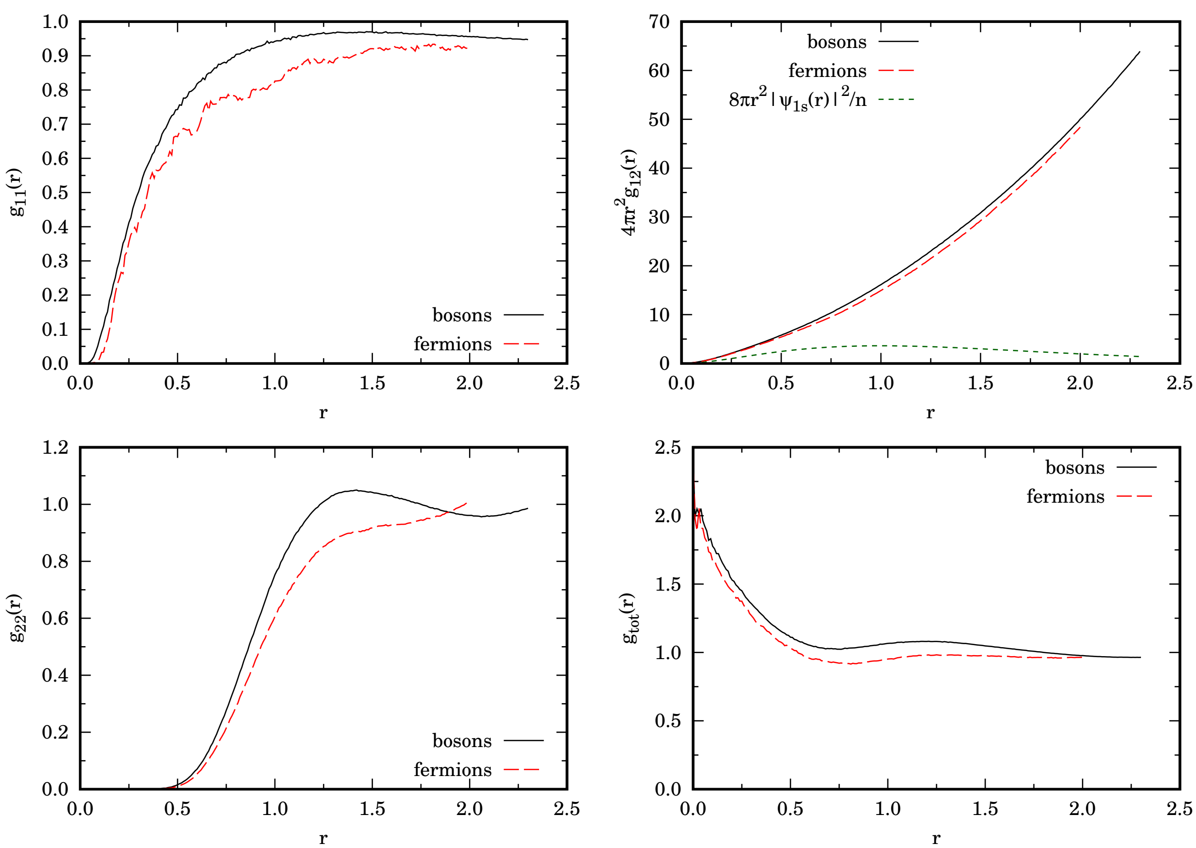

FIG. 3. We show the partial and total RDF of the H fluid for cases A$\mathscr{B}$ and B$\mathscr{B}$ in Table I where $\tau = 0.333$. In the panel for the unlike radial distribution function we also show the ground state probability distribution of the H atom $|\psi_{1s}|^2$ for comparison. The distance r is in units of a Bohr radius.

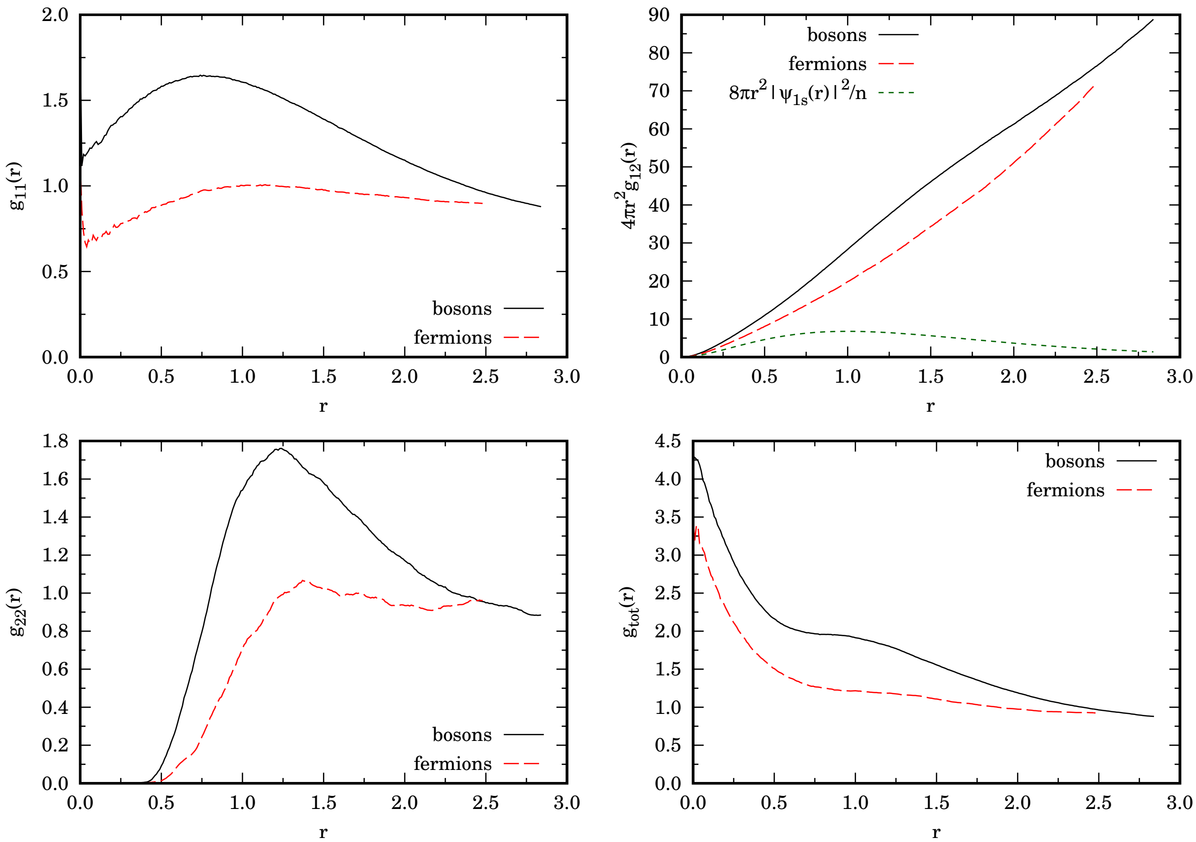

FIG. 4. We show the partial and total RDF of the H fluid for cases C$\mathscr{A}$ and D$\mathscr{B}$ in Table I where $\tau = 0.333$. In the panel for the unlike radial distribution function we also show the ground state probability distribution of the H atom $|\psi_{1s}|^2$ for comparison. The distance r is in units of a Bohr radius.

As we can see, $g_{11}(r \to 0) \to 0$ in the fermionic mixture for both cases $\mathscr{A}$ and $\mathscr{B}$ but in the bosonic mixture only for case $\mathscr{B}$. Moreover the extent of the exchange-correlation hole for the electrons is larger in the $\mathscr{B}$ case than the $\mathscr{A}$ case. With the inter-action $\mathscr{A}$ we found that the bosonic mixture reaches thermal equilibrium in a 'blob condensate' with a g_{22} pronounced peak centered at $r \approx 0.75$. In the fermionic case the mixture is much more homogeneous with a shallow incipient H_2 peak in g_{22} at $r \approx 1.5$. The large value for $g_{12}(r \to 0)$ signals the Hydrogen atoms formation. This is confirmed by the peak at $r \approx 1$ in $4\pi r^2 g_{12}(r)$. The peaks are much reduced in the $\mathscr{B}$ case. The H_2 molecule formation is made possible by the effect of the interstitial electrons between two protons which pull the two protons together when the Coulomb forces from the other charges around the two protons are screened. There will be a balance between the repulsion between the two protons and their attraction due to the interstitial electrons which determines the peak in g_{22}. In the $\mathscr{A}$ bosonic blob phase the like g_{11} RDF does not go to zero at contact but it rather has a large value reflecting the compression of the interstitial electrons and their tendency to like themselves when treated as bosons. In Fig. 8 we show snapshots of the simulations for the bosons and fermions $\mathscr{A}$ cases at equilibrium. As we can see, only the bosonic mixture forms a condensed blob of protons held together by the electrons 'glue'. The realistic fermionic mixture is much more homogeneous than the bosonic one due to Pauli exclusion principle. In Fig. 9 we show the effect, on the total potential energy, of the abrupt transition to the blob 'phase' for the bosonic mixture of case E$\mathscr{A}$ and compare it with the much smoother transition of case E$\mathscr{B}$. We also show the effect of getting closer to the continuum limit choosing $\tau = 0.1666$ for $M = 200$. Which results in a higher potential energy at equilibrium. Once again our fermionic simulations agree well with the work of Magro et al. [17]. They fix the timestep to $\tau = 0.316$ which is close to our. But respect to these authors we did not correct the long range part of the inter-action so to take into account the interaction with the periodic images outside the simulation box. This could be accomplished with the Ewald summation method or its refinements [15]. Moreover they use $N = 60$, three times larger than our.

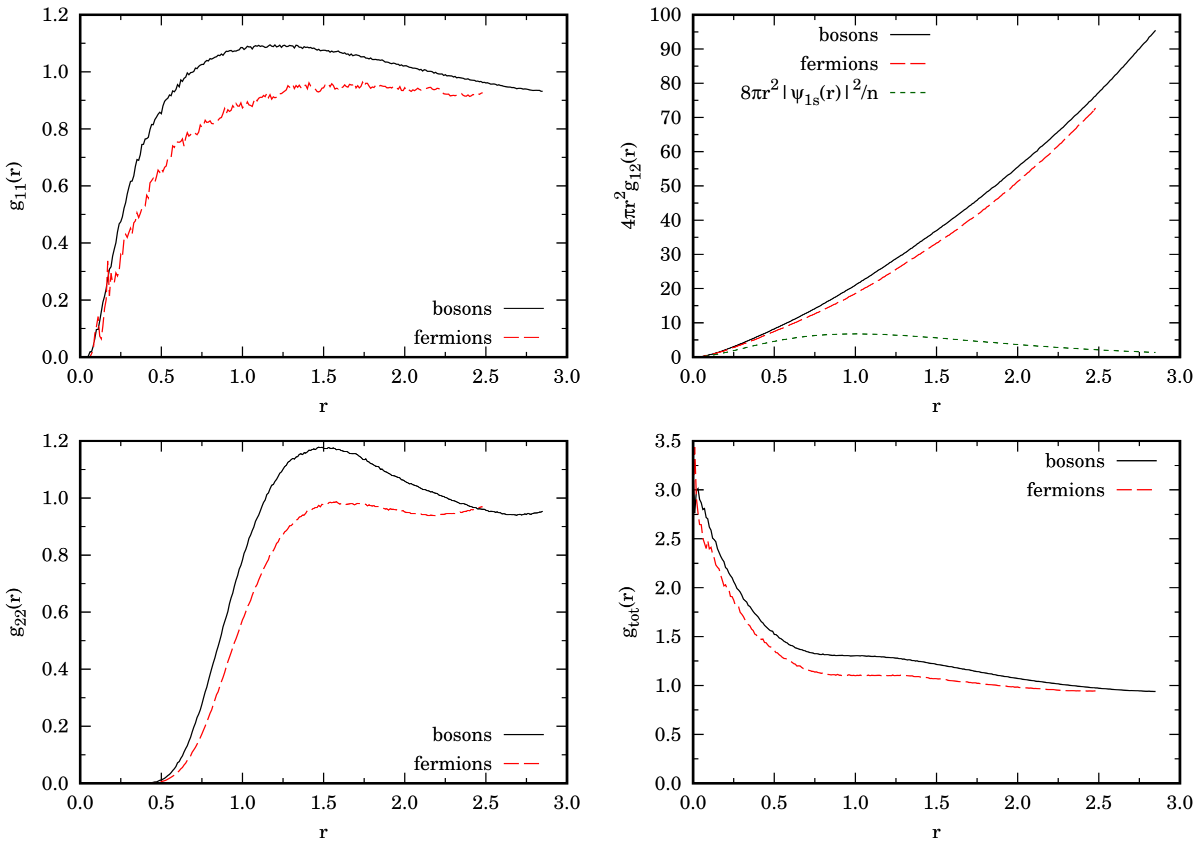

FIG. 5. We show the partial and total RDF of the H fluid for cases $C\mathscr{B}$ and $D\mathscr{B}$ in Table I where $\tau = 0.333$. In the panel for the unlike radial distribution function we also show the ground state probability distribution of the H atom $|\psi_{1s}|^2$ for comparison. The distance r is in units of a Bohr radius.

III. CONCLUSIONS

In summary, we studied the *ep* binary mixture of polarized electrons and protons through (R)PIMC to extract thermodynamic and structural properties of the plasma in thermal equilibrium. In order to approach the continuum limit with less number of timeslices we used the pair-product Coulomb inter-action within the end-point approximation.

When studying the real *ep* mixture which requires to treat electrons and protons as fermions we use the RPIMC approximate method with a free particles restriction in order to overcome the 'fermions sign problem'. This is necessary to reduce the complexity of the algorithm from exponential in the number of fermions to polynomial [10, 18]. We always compared the realistic fermionic mixture with the mixture obtained treating the particles as bosons for which the PIMC is exact and polynomial in the number of bosons. Our results for the fermionic mixture compare well with the ones of Magro et al. [17].

We explicitly compared the difference in treating the inter-action either with the end-point approximation for the pair Coulomb density matrix on both like and unlike components and using it only on the unlike component keeping the primitive approximation for the like components. As expected the latter treatment produces mixtures with less negative potential energies. Moreover, for certain thermodynamic states, the former approach may fail to reproduce the desired $g_{ee}(r \to 0) \to 0$ behavior for both the bosonic and fermionic mixture.

Our radial distribution function should be compared with Fig. 1 of Magro et al. [17]. In Ref. [4] Bonitz et al. report some evidence for the H_2 molecule formation at $T = 1200\text{K}/\mathscr{T} \approx 3.80017 \times 10^{-3}$ and $r_s = 1.44$ which means $n \approx 0.1599$, using the Coupled Electron-Ions PIMC (CEIMC). See for example their Fig. 16. It is interesting to compare the CEIMC method with our method where the mixture is treated as a whole with no need to separate the simulation of the electrons from the one of the protons as two separate Jellium [19–22] systems.

As we lower the temperature we also need to increase the number of timeslices in the path integral discretization keeping the timestep constant. If we are too far from the continuum $\tau \to 0$ limit the bosonic mixture may evolve into a more compact 'blob phase' which fills only partially the simulation box. If we call ℓ the linear dimension of the blob

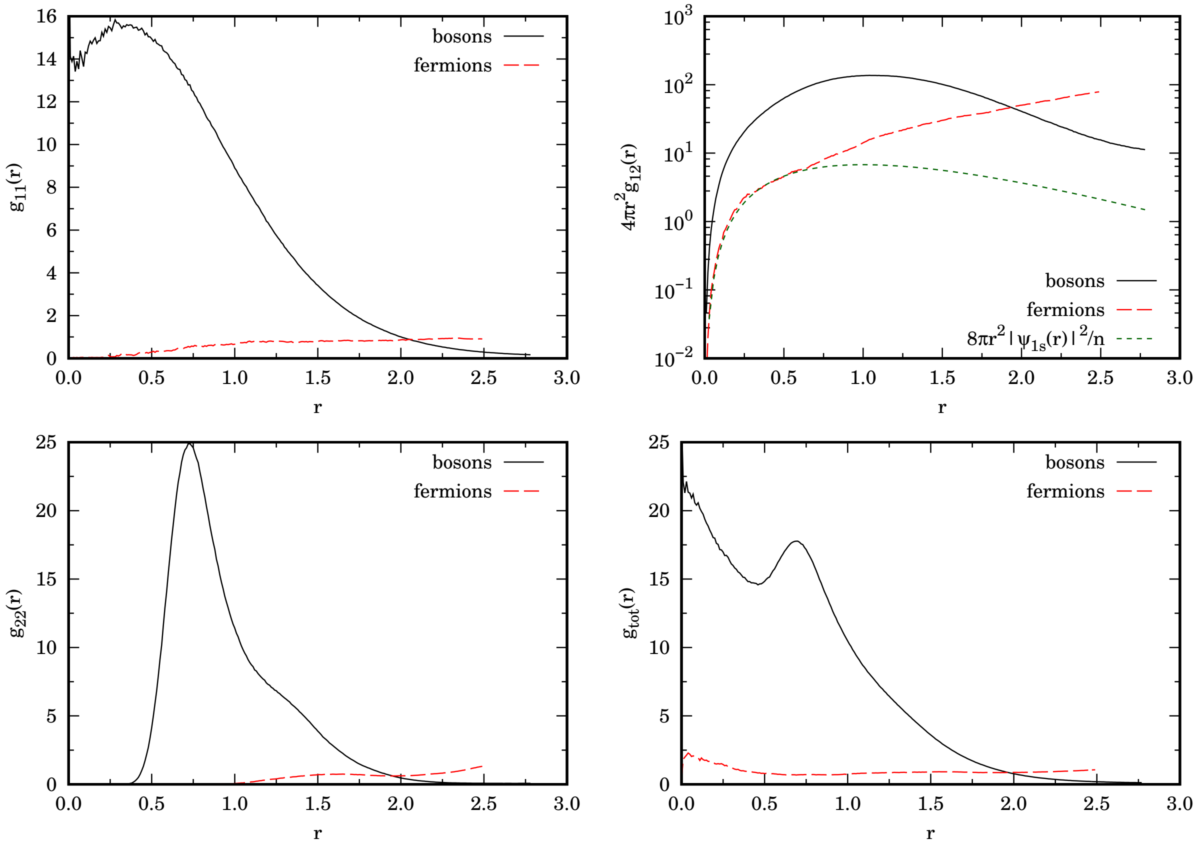

FIG. 6. We show the partial and total RDF of the H fluid for cases E$\mathscr{A}$ and F$\mathscr{A}$ in Table I where $\tau = 0.3$. In the panel for the unlike radial distribution function we also show the ground state probability distribution of the H atom $|\psi_{1s}|^2$ for comparison. They both have a maximum at about one Bohr radius as expected. The distance r is in units of a Bohr radius.

one needs to choose a number density $n \to n(L/\ell)^3$ so to fill the simulation box. Our results clearly show how the approach to the continuum $\tau \to 0$ is crucial to capture the incipience of the molecular phase. We also show that using the pair-product Coulomb action both on the like and unlike components tends to produce more compact bosonic condensate blobs, at higher temperature and density, than using it only on the unlike component and keeping the primitive action for the like components. On the other hand, on the fermionic mixture, the former approach produces more homogeneous systems than the latter, as expected for a less interacting system.

Appendix A: Description of our PIMC and RPIMC algorithms

The PIMC for bosons has two kinds of moves: a single slice move (`DISPLACE`) and a multislice move [5] (`BRIDGE`). In the single slice move we perform a uniform displacement of a single particle coordinate at a given timeslice. We accept or reject the move according to the Metropolis algorithm [3, 11]. In the multislice move we create a Brownian bridge [3] between the initial position taken from the coordinates of a particles α_a at a timeslice k_a and the final positions taken from the coordinates of a particle α_b at a timeslice $k_b = k_a + \kappa$ with κ chosen at random. Again we accept or reject the move according to Metropolis algorithm. We sample the sum over the permutations by creating two `BRIDGE` moves, so to connect particle α at the initial timeslice to particle γ at the final timeslice with one bridge and particle γ at the initial timeslice with particle α at the final timeslice. If the move is accepted one creates an exchange of two particles. We call this third move a `SWAP` move. Since any permutation of $N/2$ particles can be obtained by composing a finite number of like particles exchanges this is sufficient to simulate Bose-Einstein statistics in an exact numerical way. Using only `DISPLACE` or `BRIDGE` moves between the same particle $\alpha_a = \alpha_b$ we simulates distinguishable particles. Switching on the `SWAP` move we simulate bosonic particles. We give a weight to the frequency of each one of the three moves [3]. A MC step is made of M `DISPLACE` moves for each timeslice of the chosen particle path, a `BRIDGE` between the same particle, and a `SWAP` move. With the three moves taken with their respective frequencies.

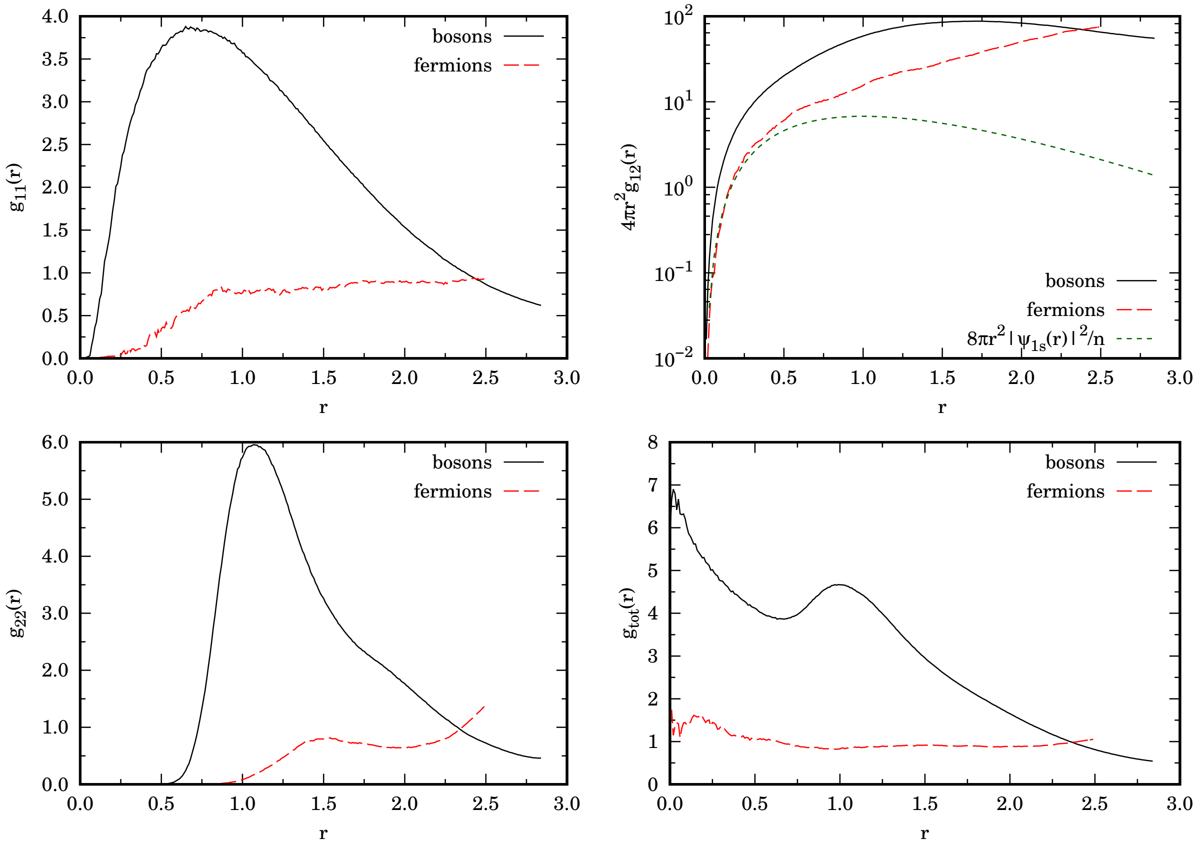

FIG. 7. We show the partial and total RDF of the H fluid for cases E$\mathscr{B}$ and F$\mathscr{B}$ in Table I where $\tau = 0.3$. In the panel for the unlike radial distribution function we also show the ground state probability distribution of the H atom $|\psi_{1s}|^2$ for comparison. The distance r is in units of a Bohr radius.

The RPIMC algorithm is the same as the PIMC for bosons decribed above choosing a nonzero frequency of the `SWAP` move and with the addition of the restriction on the paths after each MC move: given the order $N/2$ determinant

$$D_i^0(k_1,k_2) = \det\left|\left|(4\pi\lambda_i|k_1-k_2|\tau)^{-3/2}e^{-(\boldsymbol{r}_{\alpha,k_1}-\boldsymbol{r}_{\beta,k_2})^2/4\lambda_i|k_1-k_2|\tau}\right|\right|_{\alpha\beta}, \tag{A1}$$

such that $i_\alpha = i_\beta = i$, we reject the move whenever it produces a change of sign in the product $D_1^0(k_0,k)D_2^0(k_0,k)$ calculated between a reference timeslice k_r (which may be chosen at random) and any other timeslice $k \neq k_r$ [9]. Of course this method is exact only for free fermions since we impose a free particles restriction. So for our *ep* mixture is just an approximation since the particles interact with the Coulomb pair potential. Of course the enforcement of the restriction is very time consuming since one needs to calculate M order $N/2$ determinants at each of the three MC move. On the diagonal any density matrix is positive and on the path restriction $\rho(R',R;\beta) > 0$, then only even permutations, those with $\mathrm{sgn}(\mathscr{P}) = +1$, are allowed, since $\rho(\mathscr{P}R,R;\beta) = \mathrm{sgn}(\mathscr{P})\rho(R,R;\beta)$. In order to sample all even permutations it is necessary to add to our `SWAP` move a `CYCLE3` move which realizes a cyclic permutations of 3 particles.

AUTHOR DECLARATIONS

Conflicts of interest

None declared.

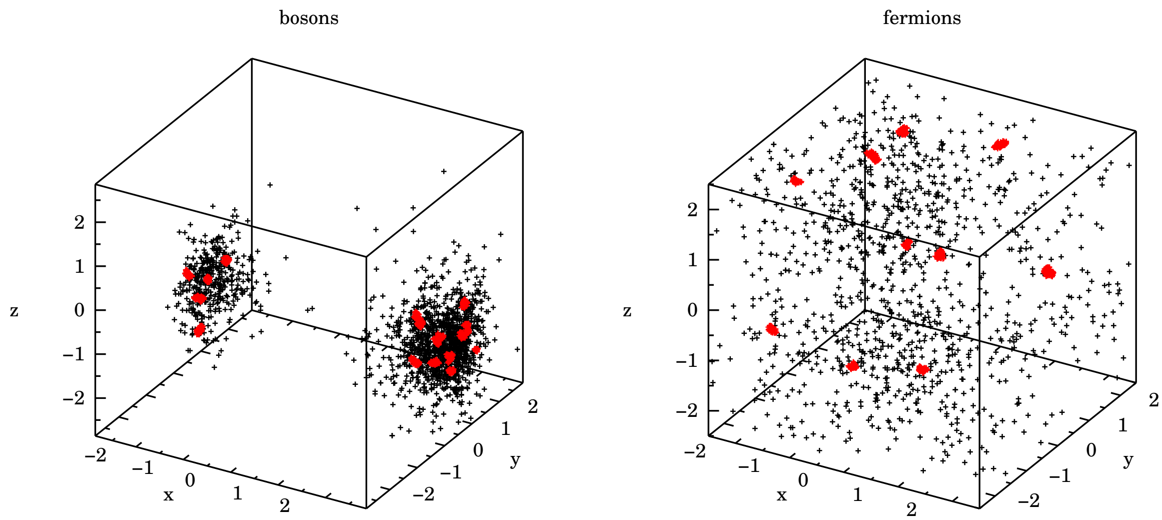

FIG. 8. Snapshots of the simulations for cases E$\mathscr{A}$ (bosons) and F$\mathscr{A}$ (fermions) at equilibrium. In these cases $\tau = 0.3$. Red points are for the heavy slow protons and black points are for the light fast electrons. Each point represents the coordinates of the given species at each timeslice along the path. Unlike the fermionic mixture the bosonic one evolves towards a 'blob condensate' which fills only partially the simulation box. Distances are in units of a Bohr radius.

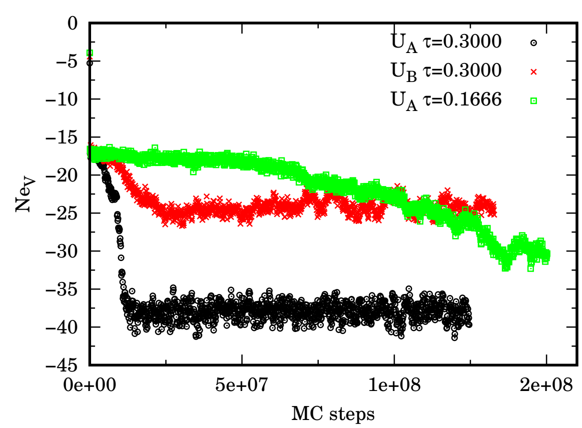

FIG. 9. We show the evolution of the potential energy Ne_V during the simulation of case E$\mathscr{A}$ ($U_{\mathscr{A}}$ $\tau = 0.3$) and E$\mathscr{B}$ ($U_{\mathscr{B}}$ $\tau = 0.3$) and for the same bosonic mixture treated with the inter-action $\mathscr{A}$ with a smaller timestep $\tau = 0.1666$ for $M = 200$. For case E$\mathscr{A}$ with $\tau = 0.3$ we see an abrupt 'transition' at about 10^7 MC steps when the blob shown in Fig. 8 is formed. The energy is in units of a Hartree.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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[1] A. Lenard, Exact Statistical Mechanics of a One Dimensional System with Coulomb Forces, J. Math. Phys. **2**, 682 (1961).
[2] S. F. Edwards and A. Lenard, Exact Statistical Mechanics of a One Dimensional System with Coulomb Forces. II. The Method of Functional Integration, J. Math. Phys. **3**, 778 (1961).
[3] M. H. Kalos and P. A. Whitlock, *Monte Carlo Methods* (John Wiley & Sons Inc., New York, 1986).
[4] M. Bonitz et al., Towards first principles-based simulations of dense hydrogen, Physics of Plasma **31**, 110501 (2024).
[5] D. M. Ceperley, Path integrals in the theory of condensed Helium, Rev. Mod. Phys. **67**, 279 (1995).
[6] R. Fantoni and G. Pastore, Monte Carlo simulation of the nonadditive restricted primitive model of ionic fluids: Phase diagram and clustering, Phys. Rev. E **87**, 052303 (2013).
[7] J. C. Kimball, Short-Range Correlations and Electron-Gas Response Functions, Phys. Rev. A **7**, 1648 (1973).
[8] E. L. Pollock, Properties and computation of the Coulomb pair density matrix, Comput. Phys. Commun. **52**, 49 (1988).
[9] D. M. Ceperley, Fermion Nodes, J. Stat. Phys. **63**, 1237 (1991).
[10] D. M. Ceperley, Path integral Monte Carlo methods for fermions, in *Monte Carlo and Molecular Dynamics of Condensed Matter Systems*, edited by K. Binder and G. Ciccotti (Editrice Compositori, Bologna, Italy, 1996).
[11] N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. M. Teller, and E. Teller, Equation of state calculations by fast computing machines, J. Chem. Phys. **1087**, 21 (1953).
[12] J. M. McMahon, M. A. Morales, C. Pierleoni, and D. M. Ceperley, The properties of hydrogen and helium under extreme conditions, Rev. Mod. Phys. **84**, 1607 (2012).
[13] D. Saumon and T. Guillot, Shock compression of deuterium and the interiors of Jupiter and Saturn, Astrophys. J. **609**, 1170 (2004).
[14] R. J. Hemley and H. K. Mao, Phase transition in solid molecular hydrogen at ultrahigh pressures, Phys. Rev. Lett. **61**, 857 (1988).
[15] V. Natoli and D. M. Ceperley, An Optimized Method for Treating Long-Range Potentials, Comput. Phys. **117**, 171 (1995).
[16] M. P. Allen and D. J. Tildesley, *Computer Simulation of Liquids* (Oxford University Press, Oxford, 1987).
[17] W. Magro, B. Militzer, D. Ceperley, B. Bernu, and C. Pierleoni, Restricted Path Integral Monte Carlo Calculations of Hot, Dense Hydrogen, in *Strongly Coupled Coulomb Systems*, edited by G. J. Kalman, J. M. Rommel, and K. Blagoev (Plenum Press, New York NY, 1998).
[18] D. M. Ceperley, Microscopic Simulations in Physics, Rev. Mod. Phys. **71**, S438 (1999).
[19] N. H. March and M. P. Tosi, *Coulomb Liquids* (Academic Press Inc Ltd, London, 1984).
[20] R. Fantoni, Jellium at finite temperature, Mol. Phys. **120**, 4 (2021).
[21] R. Fantoni, Jellium at finite temperature using the restricted worm algorithm, Eur. Phys. J. B **94**, 63 (2021).
[22] R. Fantoni, Radial distribution function in a diffusion Monte Carlo simulation of a Fermion fluid between the ideal gas and the Jellium model, Eur. Phys. J. B **86**, 286 (2013).